

CLOUD COMPUTING

Laboratory Instruction Guide

Computer Science related undergraduate programs

Three Practical Areas — 14 Lab Sessions

Area 1: Virtualization and Virtual Machines (Labs 1 – 4)

Area 2: Containers and Microservices (Labs 5 – 9)

Area 3: Private Cloud with OpenStack (Labs 10 – 14)

Based on the official OpenStack Install Guide (2024.2 Dalmatian)

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Introduction

This Laboratory Instruction Guide covers three interconnected areas of cloud computing practice for undergraduate students. Labs are numbered sequentially from 1 to 14, each building on skills from preceding sessions.

Area	Labs	Topic	Key Skills
Area 1	1 – 4	Virtualization and Virtual Machines	VirtualBox, dual-adapter setup, VM config, networking, snapshots, migration
Area 2	5 – 9	Containers and Microservices	Docker, Dockerfile, Compose, Kubernetes, persistent storage
Area 3	10 – 14	Private Cloud with OpenStack	Controller node, Compute node, Keystone, Glance, Nova, Neutron, Horizon

Hardware

Each workstation requires at least 8 GB RAM (16 GB recommended), 4 CPU cores with VT-x/AMD-V enabled in BIOS, 100 GB free disk, and a wired LAN connection. Area 3 requires two machines (physical or VMs) on the same network — one Controller and one Compute node.

OS & Prompts

All labs use Ubuntu 22.04 LTS. Commands shown with \$ may be run by any user. Commands shown with # require root or sudo. Interface names (enp0s3, enp0s8) are examples — always check your actual interface name with: `ip link show`

Standard Adapter Setup for Area 1 Labs

Every VM in this course uses TWO network adapters:

Adapter 1 (NAT): Provides internet access for package downloads. VirtualBox assigns 10.0.2.15.

The host CANNOT ping or SSH into the VM through this adapter — this is normal.

Adapter 2 (Host-Only): Provides a reachable IP (e.g. 192.168.56.x) for host-to-VM and VM-to-VM communication. Always use this IP when ping or SSH-ing from the host.

Set this up in Settings > Network before starting any VM. Both adapters must be configured.

Area 1: Virtualization and Virtual Machines

Virtualization is the foundation of all cloud computing. A hypervisor abstracts physical hardware so multiple operating system instances — called virtual machines (VMs) — can run in isolated environments on a single physical host. Area 1 uses VirtualBox, a Type 2 hosted hypervisor.

Concept	Description
Type 2 Hypervisor	Runs on top of a host OS. Examples: VirtualBox, VMware Workstation. Hardware access is mediated through the host OS.
NAT Adapter	VM gets internet via the host. VirtualBox assigns 10.0.2.15. The host CANNOT reach the VM on this IP.
Host-Only Adapter	Creates a private network between the host and VMs. The host CAN reach the VM on this IP. No internet from this adapter alone.
Snapshot	A point-in-time capture of a VM's disk and optionally RAM, enabling fast rollback.
OVA	Open Virtualisation Appliance — a portable archive of a VM's disk and configuration.

Lab 1: Installing VirtualBox, Creating a VM, and Building the Golden Image

Duration: 3 hours

Objectives

- Install VirtualBox and the Extension Pack.
- Configure a Host-Only network adapter for host-to-VM connectivity.
- Create a Ubuntu 22.04 Server VM with NAT and Host-Only adapters.
- Complete the Ubuntu installation and verify connectivity from the host.
- Update the system and install all essential packages used across the course.
- Disable cloud-init network management and prepare the VM as a golden image for cloning.

1.1 Install VirtualBox

1. Download VirtualBox 7.x from <https://www.virtualbox.org/wiki/Downloads> for your host OS (Windows, macOS, or Linux).
2. Run the installer and accept all defaults. Also install the VirtualBox Extension Pack when prompted — this adds USB 3.0, RDP, and disk encryption support.
3. Launch VirtualBox and confirm the main window opens without errors.

1.2 Create the Host-Only Network

This step must be done once. It creates the private network that allows the host machine to communicate with VMs.

4. In VirtualBox go to File > Host Network Manager (Windows/Linux) or VirtualBox > Preferences > Network (macOS).

5. Click the + icon to create a new Host-Only adapter. It will be named vboxnet0 (Linux/macOS) or VirtualBox Host-Only Ethernet Adapter (Windows).
6. Set the adapter IP to 192.168.56.1, mask 255.255.255.0. Leave the DHCP Server tab enabled with these settings: Server Address 192.168.56.100, Lower bound 192.168.56.101, Upper bound 192.168.56.254.
7. Click OK to save.

1.3 Download the Ubuntu 22.04 Server ISO

8. Download the Ubuntu Server 22.04 LTS ISO from <https://ubuntu.com/download/server> (approximately 1.4 GB). Save it to a dedicated folder such as ~/ISOs/.

1.4 Create the Virtual Machine

9. In VirtualBox click New. Set the name to Ubuntu-Lab1, Type to Linux, Version to Ubuntu (64-bit). Click Next.
10. Set memory to 2048 MB (2 GB). Click Next.
11. Select Create a virtual hard disk now, choose VDI, select Dynamically allocated, set size to 20 GB. Click Create.

1.5 Configure Both Network Adapters

WARNING

Do this before starting the VM for the first time. The default single NAT adapter gives the VM internet access but is unreachable from the host. You need both adapters.

12. Select Ubuntu-Lab1 and open Settings > Network.
13. Adapter 1 tab: confirm it is set to NAT. This is the default and gives the VM internet access.
14. Adapter 2 tab: tick Enable Network Adapter. Set Attached to: Host-only Adapter. Select your vboxnet0 adapter from the Name dropdown. Click OK.

1.6 Attach the ISO and Install Ubuntu

15. Open Settings > Storage. Under Controller: IDE click the empty disc icon. Click the disc icon beside Optical Drive on the right and select the Ubuntu ISO.
16. Go to Settings > System > Boot Order. Ensure Optical is above Hard Disk. Click OK.
17. Start the VM. Follow the Ubuntu Server installer:
 - Accept default partition layout.
 - Set a username (e.g. student) and a memorable password.
 - On the SSH Setup screen: select Install OpenSSH server. This is required for SSH access in later labs.
 - Proceed through remaining screens and confirm the installation.
18. After installation the VM reboots. Remove the ISO: Settings > Storage, click the ISO entry, choose Remove Disk from Virtual Drive. Click OK.

1.7 First Boot and Interface Identification

19. Start the VM and log in at the console with the username and password you set.

20. Identify your network interfaces:

```
ip link show
```

You should see three interfaces: lo (loopback), and two named interfaces such as enp0s3 (NAT, Adapter 1) and enp0s8 (Host-Only, Adapter 2). The exact names depend on your hardware — record them now.

21. Check the IP addresses:

```
ip addr show
```

enp0s3 will show 10.0.2.15/24 — this is the NAT address. enp0s8 will show 192.168.56.x/24 (assigned by the Host-Only DHCP server) — this is the address you use for host-to-VM access.

1.8 Verify Host-to-VM Connectivity

22. Record the enp0s8 IP address shown in the previous step (e.g. 192.168.56.101).

23. From your host machine open a terminal and ping the Host-Only IP:

```
ping 192.168.56.101 # use your actual enp0s8 IP
```

24. Confirm you receive replies. If there are no replies, check that Adapter 2 is set to Host-Only in VirtualBox settings and that the DHCP server is enabled on vboxnet0.

25. SSH into the VM from the host:

```
ssh student@192.168.56.101 # replace with your enp0s8 IP
```

26. Inside the VM, confirm internet access via the NAT adapter:

```
ping -c 4 8.8.8.8
```

NOTE

Always use the enp0s8 (Host-Only) IP when connecting from the host. The enp0s3 IP (10.0.2.15) is behind VirtualBox's NAT engine and cannot be reached from outside the VM.

1.9 System Update and Essential Package Installation

Before using the VM for any lab work, update the package lists, apply all available upgrades, and install the essential tools used across all three areas of this course. Doing this now — while the golden image is being built — means every clone inherits an up-to-date, fully equipped base.

27. Update package lists and upgrade all installed packages:

```
sudo apt update && sudo apt upgrade -y
```

28. Install core networking and diagnostic tools:

```
sudo apt install -y net-tools iproute2 iputils-ping traceroute curl wget dnsutils netcat-openbsd
```

29. Install netplan and its rendering backend (required for static IP configuration in Labs 3 and Area 3):

```
sudo apt install -y netplan.io systemd-resolved network-manager
```

30. Install system monitoring and disk utilities used in Labs 2 and 4:

```
sudo apt install -y htop sysstat iotop lsof tree nano vim
```

31. Install virtualisation guest utilities so VirtualBox can interact with the VM (clipboard sharing, dynamic screen resize, drag-and-drop):

```
sudo apt install -y virtualbox-guest-utils virtualbox-guest-additions-iso
```

NOTE

If virtualbox-guest-utils is not found in the repository, install the guest additions from the VirtualBox menu instead: Devices > Insert Guest Additions CD image, then run: `sudo mount /dev/cdrom /mnt && sudo /mnt/VBoxLinuxAdditions.run`

32. Install cloud-init tools so you can manage and suppress cloud-init behaviour cleanly (needed for Lab 3 golden image preparation):

```
sudo apt install -y cloud-init cloud-initramfs-tools
```

33. Ensure the cloud-init network configuration is disabled so it does not overwrite your netplan files on reboot. Create the disable file now:

```
sudo mkdir -p /etc/cloud/cloud.cfg.d
echo 'network: {config: disabled}' | sudo tee /etc/cloud/cloud.cfg.d/99-disable-
network-config.cfg
```

34. Install OpenSSH server if it was not selected during the Ubuntu installer (verify first):

```
sudo systemctl status ssh          # if inactive or not found, install it:
sudo apt install -y openssh-server
sudo systemctl enable ssh && sudo systemctl start ssh
```

35. Remove the cloud-init generated netplan file and replace it with your own permanent one. This is done now so every clone inherits the correct netplan setup:

```
sudo rm -f /etc/netplan/50-cloud-init.yaml
sudo nano /etc/netplan/01-lab-network.yaml
```

Add the following content:

```
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: true
    enp0s8:
      dhcp4: true
sudo chmod 600 /etc/netplan/01-lab-network.yaml
sudo netplan apply
ip addr show enp0s8          # confirm enp0s8 still has 192.168.56.x
```

36. Verify all installed tools are working:

```
htop --version
netstat -tulnp 2>/dev/null | head # from net-tools
ip route show                     # from iproute2
```

37. Clean up downloaded package files to reduce the image size before snapshotting:

```
sudo apt autoremove -y && sudo apt clean
```

38. Record the final state of the system for your lab report:

```
uname -a          # kernel version
lsb_release -a    # Ubuntu version
df -h             # disk usage
free -h           # memory
dpkg -l | wc -l   # number of installed packages
```

**Lab Report
Q1**

Explain what a Type 2 hypervisor is and how it differs from a Type 1 hypervisor. Why does VirtualBox's NAT adapter assign 10.0.2.15 to every VM, and why can the host not ping that address? What purpose does the Host-Only adapter serve? Why is it important to disable cloud-init network configuration before cloning a VM?

Lab 2: VM Configuration — CPU, Memory, Storage, and Snapshots

Duration: 2 hours

Objectives

- Adjust vCPU and memory allocation and verify the change inside the VM.
- Take and restore snapshots to understand rollback behaviour.
- Add a second virtual disk, format it, mount it, and make the mount persistent across reboots.

2.1 Adjust Hardware Settings

39. Power off Ubuntu-Lab1 cleanly:

```
sudo shutdown -h now
```

40. Open Settings > System > Processor. Set vCPUs to 2 and tick Enable PAE/NX. Click OK.

41. Start the VM, log in, and confirm the change:

```
nproc                # should now report 2
free -h              # confirm 2 GB RAM is visible
```

2.2 Take a Snapshot

42. With the VM running, go to Machine > Take Snapshot. Name it Clean-Install. Add the description: Fresh Ubuntu install, no customisations. Click OK.

43. Install a package to change the VM state:

```
sudo apt update && sudo apt install -y htop
```

44. Confirm htop works — run htop and press q to quit. Take a second snapshot named After-htop-install.

2.3 Restore a Snapshot

45. Power off the VM. In VirtualBox open Machine > Snapshots. Right-click Clean-Install and choose Restore Snapshot. Confirm when prompted.

46. Start the VM. Try to run htop — it should be reported as not found, confirming the rollback worked.

NOTE

A snapshot rollback discards everything that happened after the snapshot was taken — installed packages, created files, and configuration changes are all removed.

2.4 Add a Second Virtual Disk

47. Power off the VM. Go to Settings > Storage. Click the + icon next to Controller: SATA. Choose Add Hard Disk > Create New Disk. Select VDI, Dynamically allocated, set size to 5 GB. Name it Lab2-Disk2.vdi and click Create.

48. Start the VM. Verify the new disk appears:

```
lsblk
```

You should see sda (the primary 20 GB disk) and sdb (the new 5 GB disk). If sdb does not appear, reboot the VM.

2.5 Format, Mount, and Persist the Disk

49. Format the disk with the ext4 filesystem:

```
sudo mkfs.ext4 /dev/sdb
```

50. Create a mount point and mount the disk:

```
sudo mkdir -p /mnt/data
sudo mount /dev/sdb /mnt/data
df -h | grep sdb          # confirm mount and available space
```

51. Write a test file to confirm the disk is writable:

```
echo 'Lab 2 disk test' | sudo tee /mnt/data/testfile.txt
sudo cat /mnt/data/testfile.txt
```

52. Make the mount persistent so it survives a reboot. First get the disk's UUID:

```
sudo blkid /dev/sdb
```

Note the UUID value (a string like 6a8e3c1f-...). Now edit /etc/fstab:

```
sudo nano /etc/fstab
```

Add this line at the bottom, replacing the UUID with your actual value:

```
UUID=your-uuid-here /mnt/data ext4 defaults 0 2
```

53. Test the fstab entry before rebooting:

```
sudo umount /mnt/data
sudo mount -a          # mounts everything in fstab
df -h | grep sdb       # confirm it remounted
```

54. Reboot and verify the mount persists:

```
sudo reboot
df -h | grep sdb       # sdb should still be mounted after reboot
```

**Lab Report
Q2**

Describe what happens to installed software and created files when a snapshot is restored. What is the difference between a snapshot and a full clone? What is /etc/fstab and why is it necessary for a disk mount to survive a reboot?

Lab 3: VM Networking — NAT, Bridged, and Host-Only in Depth

Duration: 2 hours

Objectives

- Understand and demonstrate the behaviour of NAT, Bridged, and Host-Only adapter modes.
- Understand which adapter to use for which purpose.
- Prepare a golden image by removing cloud-init network files and clearing the machine-id.
- Clone the golden image and confirm each clone receives a unique IP address.

3.1 Networking Mode Reference

Mode	VM gets internet?	Host can reach VM?	VMs reach each other?	Best used for
NAT	Yes (via host)	No — host cannot reach 10.0.2.x	No	Outbound internet from any VM
Bridged	Yes (own LAN IP)	Yes	Yes (if on same LAN)	Simulating a real networked server
Host-Only	No (unless combined with NAT)	Yes — host can reach 192.168.56.x	Yes	Lab networks, host-to-VM SSH
Internal	No	No	Yes	Fully isolated inter-VM networks

NOTE

In this course every VM uses NAT (Adapter 1) + Host-Only (Adapter 2). The NAT adapter provides internet. The Host-Only adapter provides host access. This combination gives the best of both worlds.

3.2 Observe NAT Behaviour

55. With Ubuntu-Lab1 running (both adapters configured), run from inside the VM:

```
ping -c 4 8.8.8.8           # internet works via NAT
curl -s https://ifconfig.me # shows host's public IP, not VM's
ip addr show enp0s3         # shows 10.0.2.15 — the NAT address
```

56. From the host, try to ping the NAT address:

```
ping 10.0.2.15             # no reply — expected behaviour
```

This confirms that the NAT adapter provides outbound internet access only. The host cannot initiate connections to the VM through NAT.

3.3 Use the Host-Only Adapter for Host Access

57. From inside the VM, note the enp0s8 IP:

```
ip addr show enp0s8        # shows 192.168.56.x
```

58. From the host, ping the enp0s8 address:

```
ping 192.168.56.101        # replace with your actual IP — should reply
```

59. SSH from the host:

```
ssh student@192.168.56.101
```

This is the correct way to access a VM from the host throughout this course.

3.4 Observe Bridged Mode

60. Power off Ubuntu-Lab1. Change Adapter 1 from NAT to Bridged Adapter. Select your host's active physical NIC from the Name dropdown. Start the VM.

```
ip addr show enp0s3 # now shows a LAN IP (e.g. 192.168.1.x) from your router
```

61. From a separate machine on the same LAN, ping the VM's bridged IP. Confirm it replies. This shows the VM is now a visible node on the physical network.
62. Revert to NAT before the next section: power off, change Adapter 1 back to NAT, start the VM.

3.5 Finalise the Golden Image and Take a Snapshot

The cloud-init network file was already removed and 01-lab-network.yaml was already created in Lab 1 section 1.9. The only remaining step before cloning is to clear the machine identity so every clone boots with a unique DHCP client ID.

63. Verify the cloud-init network disable file is in place and 50-cloud-init.yaml is gone:

```
cat /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg # should print: network:
{config: disabled}
ls /etc/netplan/ # should show 01-lab-network.yaml only - no 50-cloud-
init.yaml
```

64. If 50-cloud-init.yaml is still present (e.g. the VM was not updated in Lab 1), remove it now:

```
sudo rm -f /etc/netplan/50-cloud-init.yaml
sudo netplan apply
```

65. Clear the machine-id. Leave the file empty — do not delete it and do not run systemd-machine-id-setup. Systemd will generate a fresh unique ID on the next boot of every clone:

```
sudo truncate -s 0 /etc/machine-id
sudo rm -f /var/lib/dbus/machine-id
cat /etc/machine-id # should be empty (zero bytes)
```

WARNING

Do NOT run systemd-machine-id-setup after truncating. Leave the file empty. If you run it now the ID gets fixed again and all clones will share it.

66. Shut down Ubuntu-Lab1 cleanly:

```
sudo shutdown -h now
```

67. Take the Golden-Image snapshot in VirtualBox before cloning:

Select Ubuntu-Lab1 in VirtualBox. Go to Machine > Take Snapshot. Name it Golden-Image. Add the description: Clean Ubuntu 22.04, all packages installed, cloud-init suppressed, machine-id cleared.

3.6 Clone and Build the Host-Only Network

68. Clone Ubuntu-Lab1 from the Golden-Image snapshot: right-click > Clone. Name it Ubuntu-Lab3-B. Select Full clone and tick Reinitialise MAC addresses of all network cards. Click Clone.
69. Verify both VMs have Adapter 1 = NAT and Adapter 2 = Host-Only (vboxnet0). Start both VMs.
70. On first boot, systemd generates a fresh machine-id on Ubuntu-Lab3-B, the Host-Only DHCP server sees a new client, and assigns a different IP. Check both:

```
# Inside Ubuntu-Lab1:
ip addr show enp0s8          # e.g. 192.168.56.101
# Inside Ubuntu-Lab3-B:
ip addr show enp0s8          # e.g. 192.168.56.102 - must be different
```

NOTE

If both VMs show the same enp0s8 IP, power off Ubuntu-Lab3-B and run: `sudo truncate -s 0 /etc/machine-id && sudo rm -f /var/lib/dbus/machine-id` — then reboot. Do not run `systemd-machine-id-setup`.

71. From Ubuntu-Lab1, ping Ubuntu-Lab3-B using its enp0s8 IP:

```
ping -c 4 192.168.56.102    # replace with actual IP of Ubuntu-Lab3-B
```

72. Install openssh-server on Ubuntu-Lab3-B if it was not selected during setup:

```
sudo apt install -y openssh-server
```

73. From Ubuntu-Lab1, SSH into Ubuntu-Lab3-B:

```
ssh student@192.168.56.102
```

**Lab Report
Q3**

Draw a network diagram showing both VMs with their two adapters, labelled with IP addresses and adapter modes. Explain why cloned VMs get the same IP when machine-id and cloud-init network files are not cleared. For each of the four adapter modes (NAT, Bridged, Host-Only, Internal), describe a real-world scenario where it is the correct choice.

Lab 4: VM Portability, Cloning, and Resource Monitoring

Duration: 2 hours

Objectives

- Export a VM as OVA and re-import it to demonstrate cold migration.
- Compare full clone and linked clone in size and independence.
- Monitor resource usage from inside the VM and from the VirtualBox host panel.

4.1 Export and Import (Cold Migration)

74. Power off Ubuntu-Lab1. Go to File > Export Appliance. Select Ubuntu-Lab1. Set format to OVA. Choose an output path and click Export. Note the file size and how long it takes.
75. Import it back: File > Import Appliance. Browse to the .ova file. Change the VM name to Ubuntu-Imported. Accept defaults and click Import.
76. Start Ubuntu-Imported. Log in and confirm the testfile from Lab 2 is still present:

```
cat /mnt/data/testfile.txt
```

The full disk and filesystem state was preserved through export and import.

4.2 Full Clone vs Linked Clone

77. Right-click Ubuntu-Lab1 > Clone. Select Full clone. Name it Ubuntu-Full-Clone. Tick Reinitialise MAC addresses. Click Clone.
78. Repeat, this time selecting Linked clone. Name it Ubuntu-Linked-Clone.
79. Compare disk sizes on the host (run from a host terminal):

```
du -sh ~/VirtualBox\ VMs/Ubuntu-Full-Clone/
du -sh ~/VirtualBox\ VMs/Ubuntu-Linked-Clone/
```

The linked clone will be several gigabytes smaller because it shares the base disk with the parent VM and only stores changes.

4.3 Resource Monitoring from the VirtualBox Host

80. Start Ubuntu-Lab1. Open Machine > Session Information (or press Host+N) to see real-time CPU, disk I/O, and network graphs.
81. Install the stress tool and generate CPU load:

```
sudo apt install -y stress
stress --cpu 2 --timeout 30
```

82. Observe the CPU graph spike in the Session Information panel during the 30 seconds.

4.4 Resource Monitoring Inside the VM

83. Install monitoring tools if not present:

```
sudo apt install -y htop sysstat
htop                                # interactive process view - q to quit
vmstat 1 10                         # CPU, memory, I/O every second, 10 samples
iostat -x 1 5                       # disk utilisation stats
```

Lab Report
Q4

What is cold migration and how does it differ from live migration? Compare the full clone and linked clone — when is each appropriate? Run three 2 GB VMs simultaneously on your 8 GB host and record the vmstat output. What happens to the idle CPU percentage?